

# GIS Intake Assistant

## Project Evolution Log (v0.0 → v0.2.16)

### Purpose

Build a drafting intake tool that reduces GIS research time by helping locate:

- GIS / Parcel Maps
- Zoning Maps
- Zoning Ordinances
- Setback References

while creating a growing internal source database for future projects.

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## STEP 1 — Initial GIS Lookup Prototype

### Goal

Avoid manually Googling GIS websites for every project.

### Features

- Address input
- City input
- State input
- GIS search button
- Geocoding using Nominatim
- County detection
- State detection

### Result

Successfully identified:

County  
State  
Formatted Address

from project addresses.

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## STEP 2 — Automated Source Discovery

### Goal

Automatically suggest likely GIS sources.

### Features

DuckDuckGo search integration.

Suggested:

GIS / Parcel Sources

based on detected jurisdiction.

### Problem

Results were inconsistent.

Example:

Correct county mixed with unrelated counties  
Random GIS sites

### Lesson

Simple keyword searches were not enough.

Jurisdiction awareness would be required.

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## STEP 3 — Google Sheets Database

### Goal

Stop repeatedly searching for the same GIS links.

### Features

Google Sheets integration.

Created:

GIS Sources  
Search Log

tables.

Users could:

Save verified GIS source  
Save zoning source  
Store notes

## Result

Application became self-learning.

Every verified source improved future lookups.

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# STEP 3A — Search Logging

## Goal

Track actual usage.

## Features

Every lookup logs:

Timestamp  
Address  
County  
State  
Saved Source Count  
Suggested Source Count

## Result

Provides future insight into:

Which jurisdictions are reused  
Where source coverage is weak

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## STEP 4 — UI / UX Overhaul

### Goal

Make the tool feel like a drafting utility rather than a developer tool.

### Major Changes

#### Layout

Moved to:

Two-column workflow

Left:

Lookup  
Copy Address  
Discovery Sources  
Save Source

Right:

Detected Area  
Saved Sources

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### Visual Style

Adopted:

IBM Plex Mono

across the entire application.

Goals:

Minimal  
Technical  
CAD-like  
Professional

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## Usability Improvements

Added:

Always-visible Saved Sources  
Copy Address section  
Improved spacing  
Cleaner section hierarchy  
Larger typography

Removed:

Clutter  
Unused space  
Overly developer-oriented layout

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## Result

Current interface became suitable for daily drafting use.

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# STEP 5 — Setback / Ordinance Discovery

## Goal

Reduce time spent searching for setback requirements.

## Initial Idea

Search directly for:

## Setbacks

### Problem

Results were poor.

Examples:

Legal blogs  
News articles  
Wrong counties  
Random PDFs

### Lesson

Setbacks rarely exist as standalone resources.

Most jurisdictions store setbacks inside:

Zoning Ordinance  
Development Code  
Land Development Code  
Municipal Code

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### Pivot

Instead of searching for setbacks directly:

Search for:

Official Zoning Ordinance  
Development Code  
Land Development Code

### Result

Much more useful for actual drafting research.

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## STEP 5A — Jurisdiction-Aware Filtering

### Goal

Prevent unrelated jurisdictions from appearing.

### Added Logic

Search results must match:

Detected City  
Detected County  
Detected State

### Result

Significantly cleaner results.

Example:

Grand Junction  
Mesa County  
Colorado

now prioritizes:

Mesa County  
Grand Junction  
Colorado

instead of nearby jurisdictions.

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## STEP 5B — Duplicate Suppression

### Goal

Avoid recommending sources already saved.

### Logic

Discovery results:

Exclude URLs already stored in Google Sheets

## Result

Saved Sources become:

Verified references

while Discovery becomes:

New candidates

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# Google Credential Incident

## What Happened

GitGuardian detected:

Google Cloud Key exposed in GitHub repository

Cause:

.streamlit/secrets.toml

was accidentally committed.

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## Initial Response

Attempted credential rotation.

Mistakenly ended up with:

Old service account  
New service account

Application still referenced the old account.



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## Symptoms

Application displayed:

```
invalid_grant  
account not found
```

while Google Sheets failed.

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## Resolution

Created new service account.

Updated:

```
Local secrets.toml  
Streamlit Cloud Secrets  
Google Sheet sharing permissions
```

Added:

```
.streamlit/secrets.toml
```

to:

```
.gitignore
```

Removed secrets from repository tracking.

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## Lesson

Google integrations require BOTH:

```
Valid credentials  
+  
Spreadsheet access permissions
```

Many authentication issues are permission-related, not code-related.

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## Current Version

v0.2.16

### Working Features

- ✓ Address lookup
- ✓ County detection
- ✓ State detection
- ✓ GIS source discovery
- ✓ Zoning map discovery
- ✓ Ordinance / setback discovery
- ✓ Google Sheets source database
- ✓ Search logging
- ✓ Saved source notes
- ✓ Duplicate suppression
- ✓ Source saving workflow
- ✓ Streamlit deployment
- ✓ IBM Plex Mono UI

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## Current Workflow

1. Enter project address
2. Detect county/state
3. Review saved sources
4. Review new discovery sources
5. Open GIS portal
6. Open zoning map

- 7. Open ordinance / setback references
  - 8. Save verified sources
  - 9. Improve database over time
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## Current Objective

### Real-World Testing

Use the application during actual Evolution Drafting projects.

Evaluate:

Does it save time?  
Does it find useful ordinance references?  
Are setback sources useful?  
Which counties still produce poor results?

The next development phase should be driven by real workflow observations rather than adding features for their own sake.

Current status:

Production Test Build  
v0.2.16